

Aryan Sultan Khan

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I am an early-career professional with a strong foundation in statistical modelling, forecasting, and predictive analytics. Proficient in R, MATLAB, SPSS, and Excel, I have worked with complex datasets to develop quantitative models for economic forecasting, optimisation, and strategic decision-making. My experience includes applying statistical methods to extract key insights, developing models to inform business decisions, and presenting impactful recommendations to senior stakeholders. I am a co-host on [Forecasting Impact](#), a podcast by the International Institute of Forecasters.

Technical Skills

- **Statistical Modelling Tools:** R, STATA, MATLAB, Python, and Power BI
- **Data Analysis tools:** MS Excel
- **Quantitative Techniques:** Time series analysis, Statistical modelling, Economic Forecasting, Predictive Modelling, Risk Analysis

Work Experience

Vital Agri Nutrients Ltd & Vital Green Ltd

Remote

Data Analyst

June 2024 – present

- Analyse and review global and local agricultural reports, including USDA (GAIN) and NFDC publications, to identify trends in demand for NPK blends and biofertilizers, influencing AUD 250K+ in production planning decisions.
- Research seasonal patterns, market demand, and crop-specific factors to provide insights supporting production planning and decision-making processes.
- Prepare and compile summary reports for internal use, helping production and sales teams understand key demand drivers and market conditions. Recommendations enhanced forecast accuracy, improving decision-making for production and sales teams.
- Use basic statistical methods in R to assist with demand forecasting, incorporating factors such as seasonal variations and external policies affecting crop demand such as the government policies, also accounting for nutrient-specific price sensitivity.
- Collaborate in weekly team meetings to review progress, align on goals, and ensure the delivery of high-quality work through effective teamwork and coordination.

Skills: Data Analysis, Statistical Modelling, R, Excel

Hudson Institute of Medical Research

Melbourne, VIC

Intern (Industry Placement)

March 2023 – June 2023

- Developed a discrete event simulation model to optimise bed allocation and utilisation in the stroke unit. The model showed that increasing beds from 7 to 11 for an annual patient load of 350 reduced percent of patients waiting to be admitted into the stroke unit to under 5%.
- Evaluated bed capacities ranging from 5 to 50 for annual patient loads between 350 and 2,500, assessing their impact on patient wait times, wait percentages, and bed utilisation.
- My analysis demonstrated that optimising bed allocation reduced patient wait times for stroke unit admission to under three hours.
- Engineered an interactive [Shiny web application](#) to visualise the effects of patient load and bed capacity on stroke ward efficiency enabling stakeholders to make informed decisions quickly.
- Collaborated closely with medical professionals to validate the simulation model, ensuring alignment with clinical realities.
- Presented project findings to senior stakeholders, effectively communicating key insights on patient outcomes.
- Received positive feedback from my supervisor and was awarded a distinction grade for my work.

Skills: Simulation Modelling, Forecasting, R programming, R-Shiny, Data visualisation, Collaboration

Merged Areas Governance Project

Remote

Research Assistant to Dr. Samina Afridi

November 2022 – February 2023

- Collaborated remotely with Dr. Samina Afridi, a UNDP-affiliated consultant, on the Merged Areas Governance Project.
- Proactively established and maintained strong networks with key stakeholders, including local authorities, to source critical socio-economic data.
- Conducted detailed data gathering from government bodies such as the Social Welfare Department and Zakat Management and Information Systems (ZMIS) to capture key metrics on vocational training, female workforce participation, and financial aid distribution.
- Utilised R to clean, analyse, and visualise data on demographics, employment trends, and educational infrastructure, producing insights to guide policy recommendations.
- Supported the development of data-driven recommendations aligned with project goals, to inform policy strategies enhancing female literacy, vocational training access, and economic opportunities in the merged districts.
- Ensured consistent, clear communication with Dr. Afridi to align all research activities with project objectives and keep the team informed on data updates and findings, facilitating the delivery of accurate and actionable insights.

Skills: Data Analysis, Quantitative & Qualitative Research, Collaboration

Education

Monash University

Master of Business Analytics

Melbourne, VIC

February 2021 - December 2023

University of London (Worldwide)

Bachelor of Science in Economics and Management

August 2015 - May 2018

Project Experience

Retail Forecasting Project

Melbourne, VIC

April 2022 - May 2022

- Led the development of data-driven economic forecasts for the monthly turnover of the pharmaceutical, cosmetic, and toiletry goods industry in New South Wales (NSW), employing advanced statistical techniques in R. (*Project available [here](#)*)
- Conducted a thorough preliminary analysis of the dataset, uncovering underlying trends, seasonality, and cyclic patterns that were critical for the development of accurate forecast models.
- Expertly applied Exponential Smoothing State Space Models (ETS) and Auto-Regressive Integrated Moving Average (ARIMA) models to generate forecasts, achieving an optimal balance between model complexity and predictive accuracy.
- Performed rigorous back-testing, evaluating the performance of ETS and ARIMA models on training data and comparing it against out-of-sample test data to ensure the robustness and reliability of the forecasts.
- Implemented a systematic model selection process, incorporating both accuracy metrics and business considerations, to identify and recommend the most suitable forecasting model, significantly enhancing the predictive accuracy and value of the forecasting exercise.

References

References are available upon request.